

Amulet Motion Controller

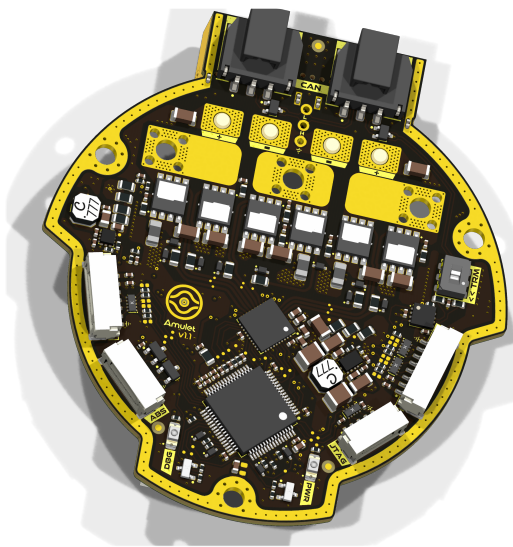
Variant:

2026-08-17

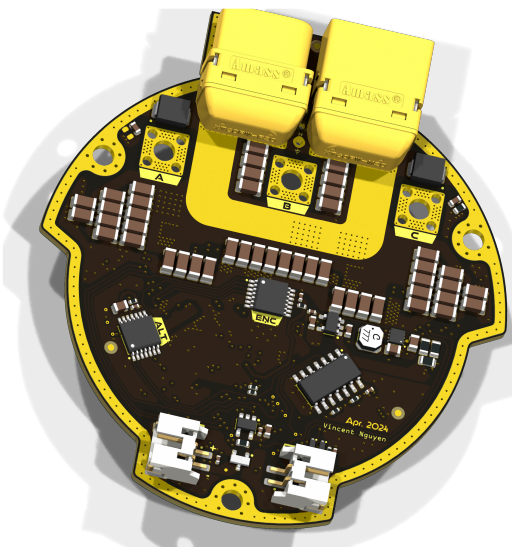
Rev 1.1.1+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

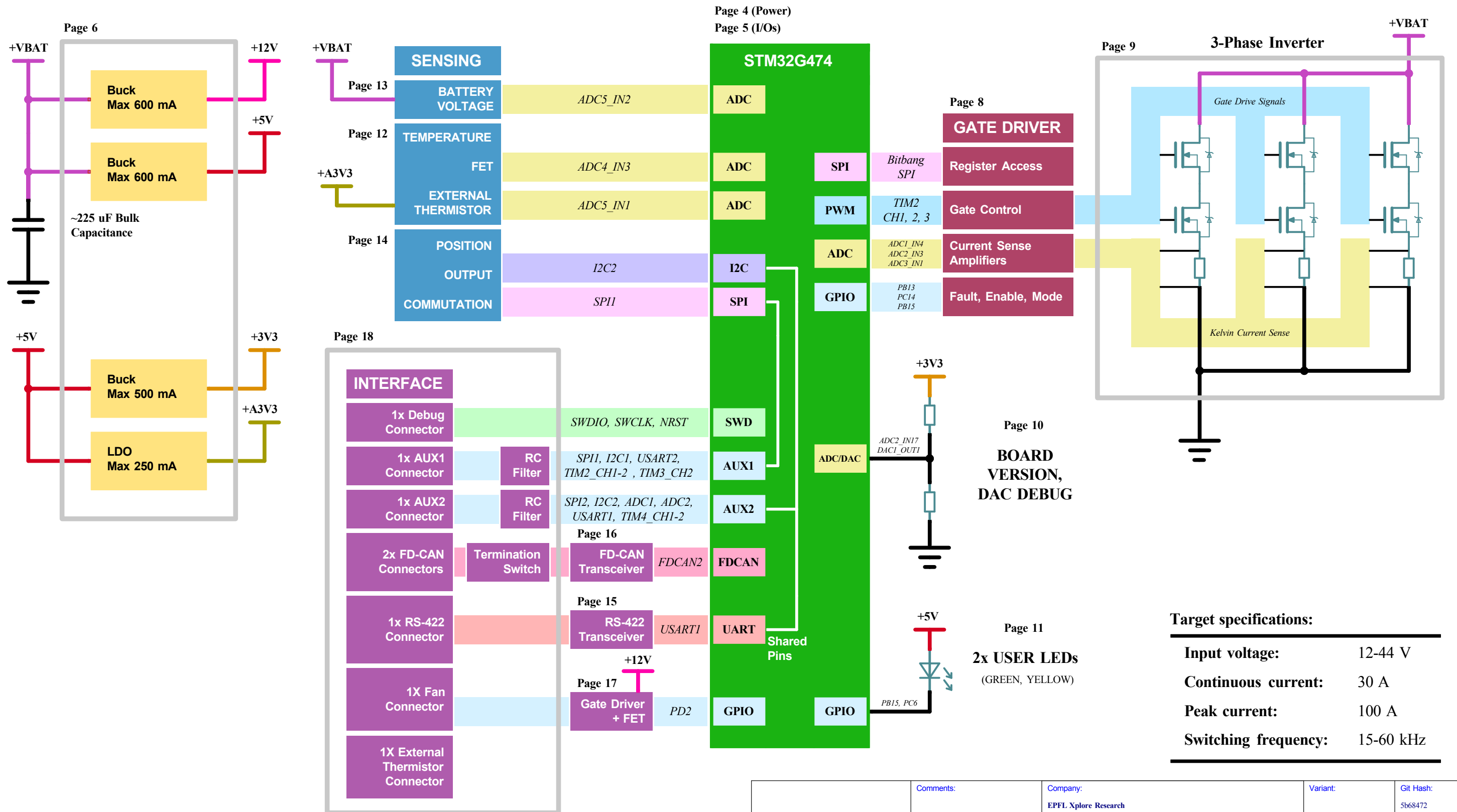
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

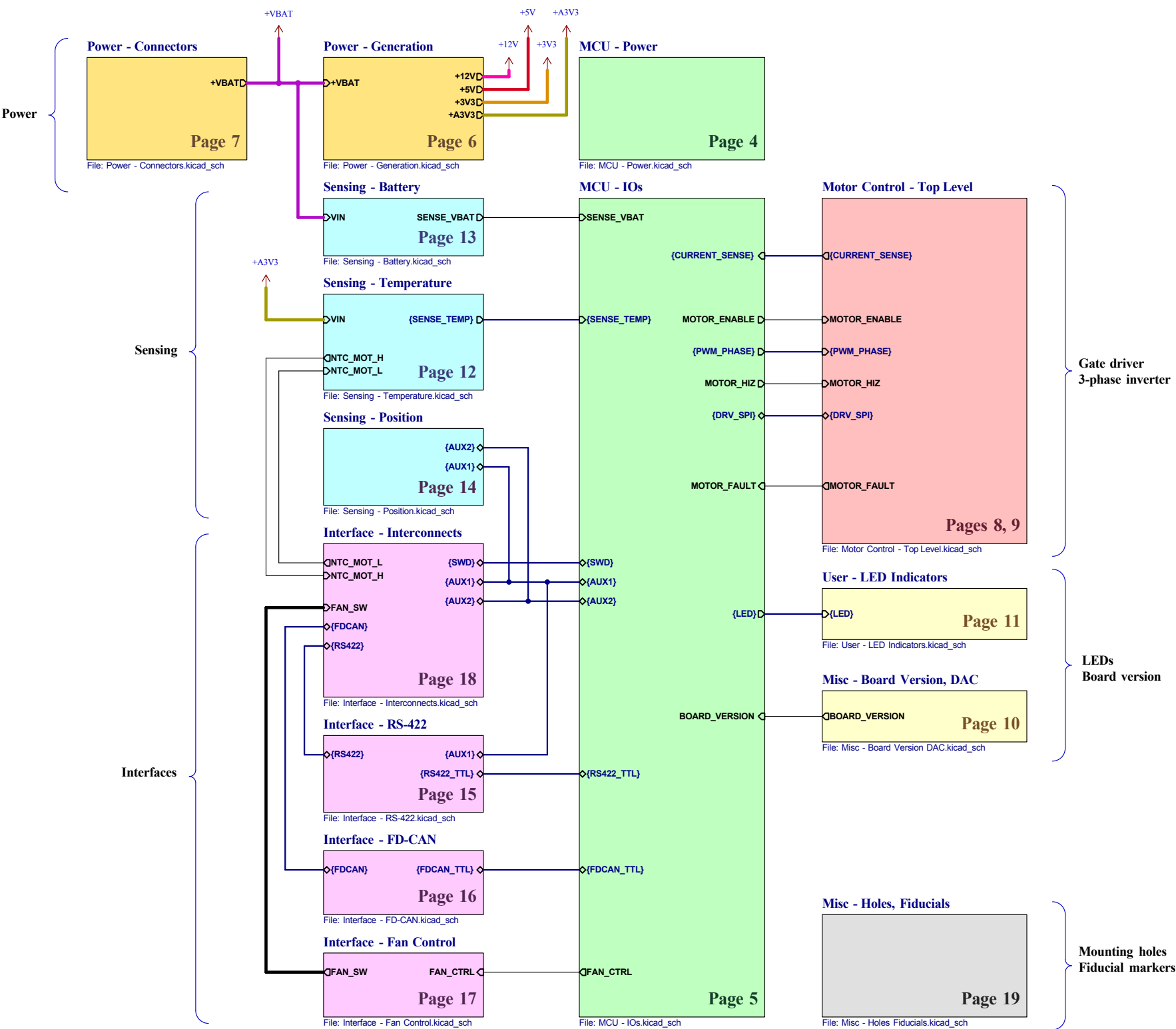
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	Sheet Title: Cover Page	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Revision: 1.1.1+ (Unreleased)	
			Size: A3	Sheet: 1 of 21

[2] Block Diagram



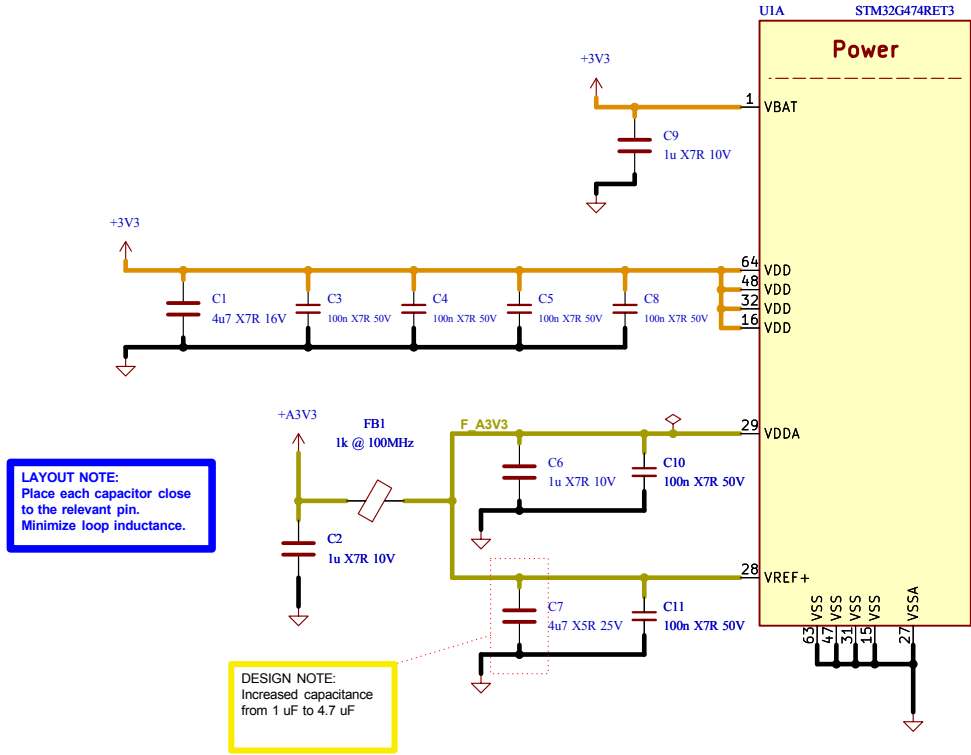
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	EPFL Xplore Research			5b68472
	Board Name:	Project Name:		
	Amulet Motion Controller	Chienpanzé		
Sheet Title:	File Name:	Designer:	Date:	Revision:
Block Diagram	Block Diagram.kicad_sch	Vincent Nguyen	2024-04-13	1.1.1+ (Unreleased)
Sheet Path:	Reviewer:	Size:	Sheet:	
/Block Diagram/		A3	2 of 21	

[3] Project Architecture



Comments:	Company:		Variant:	Git Hash:
	EPFL Xplore Research			5b68472
	Board Name:		Project Name:	
Sheet Title:	File Name:	Designer:	Date:	Revision:
Project Architecture	Project Architecture.kicad_sch	Vincent Nguyen	2023-12-22	1.1.1+ (Unreleased)
Sheet Path:	Reviewer:		Size:	Sheet:
/Project Architecture/			A3	3 of 21

[4] MCU - Power



Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Sheet Title: MCU - Power	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/MCU - Power/	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1.1+ (Unreleased)
			Reviewer:	Size: A4	Sheet: 4 of 21

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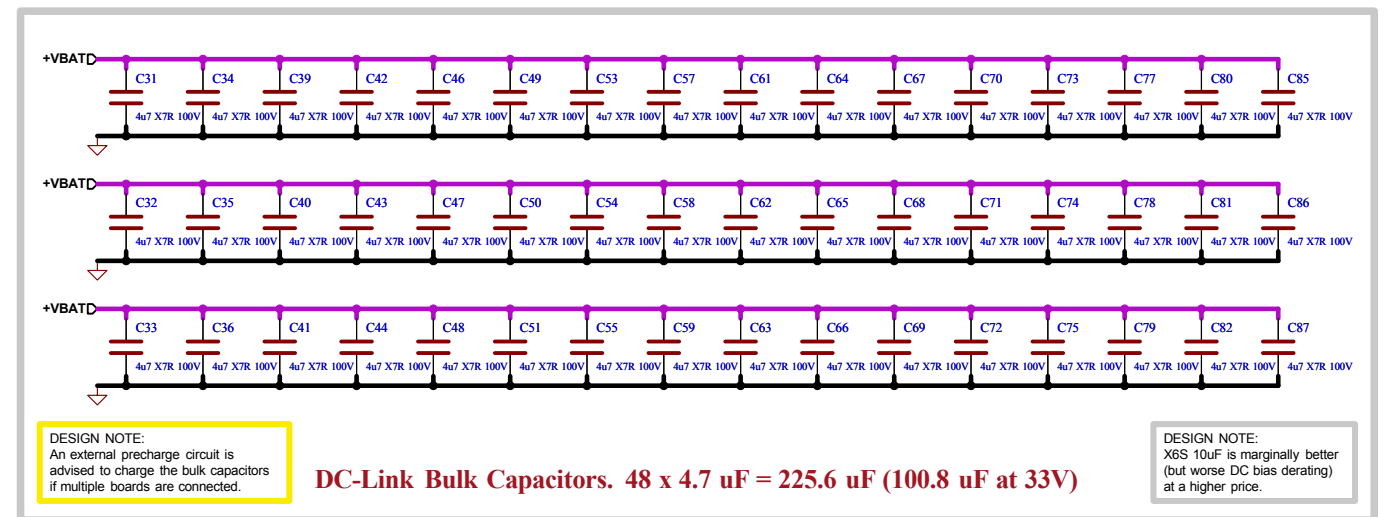
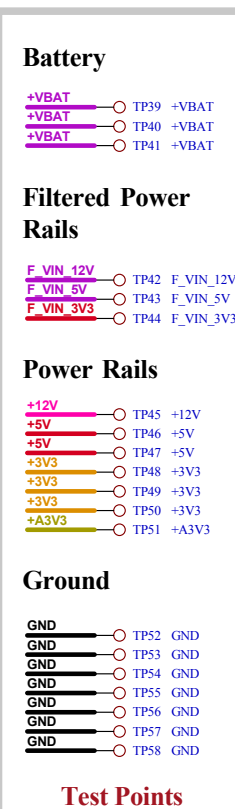
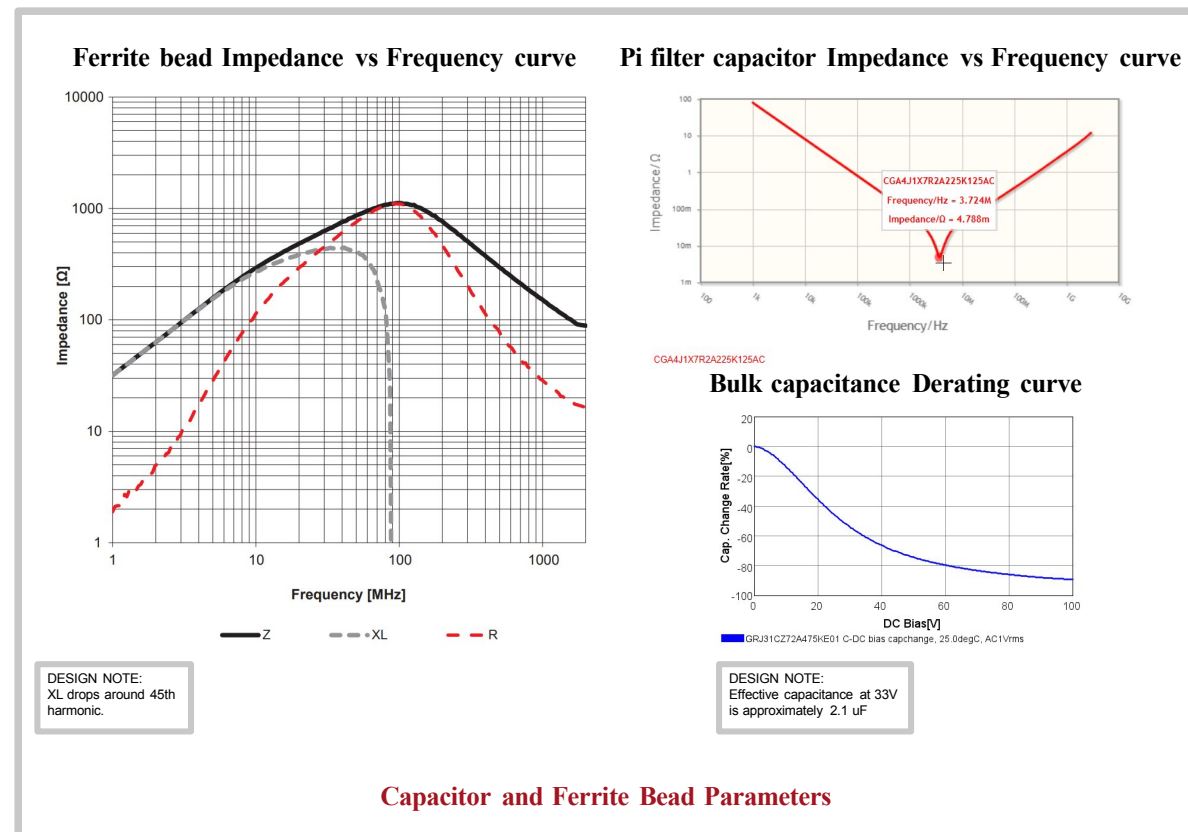
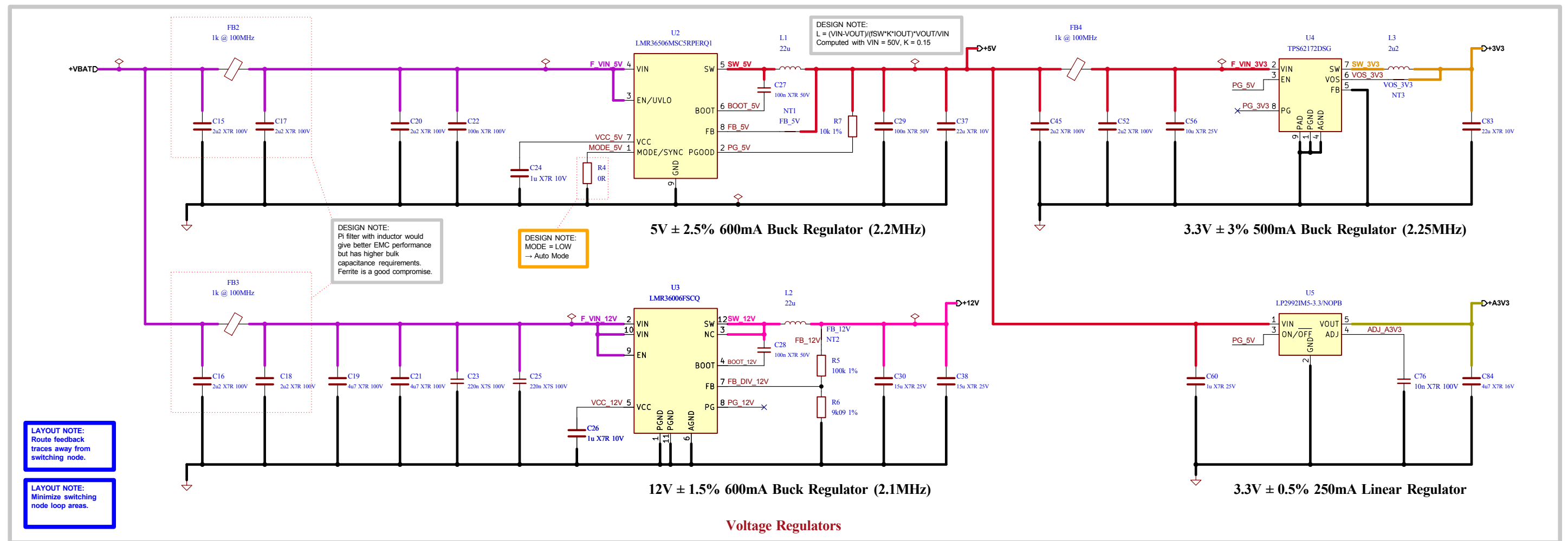
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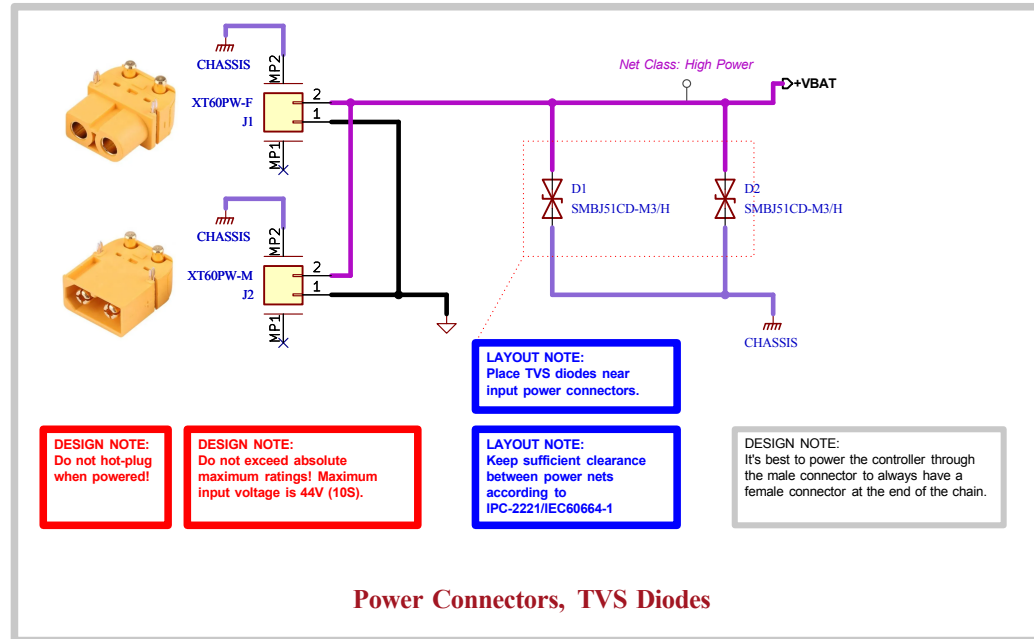
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[6] Power - Generation



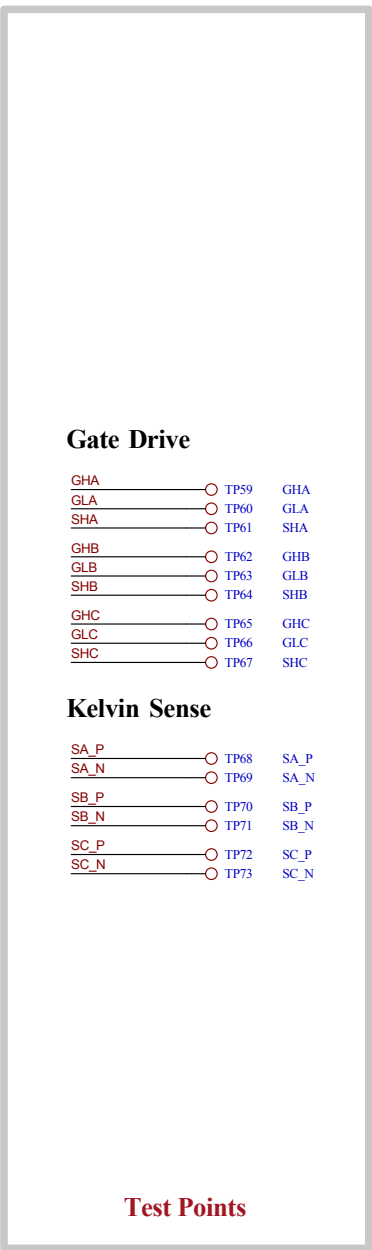
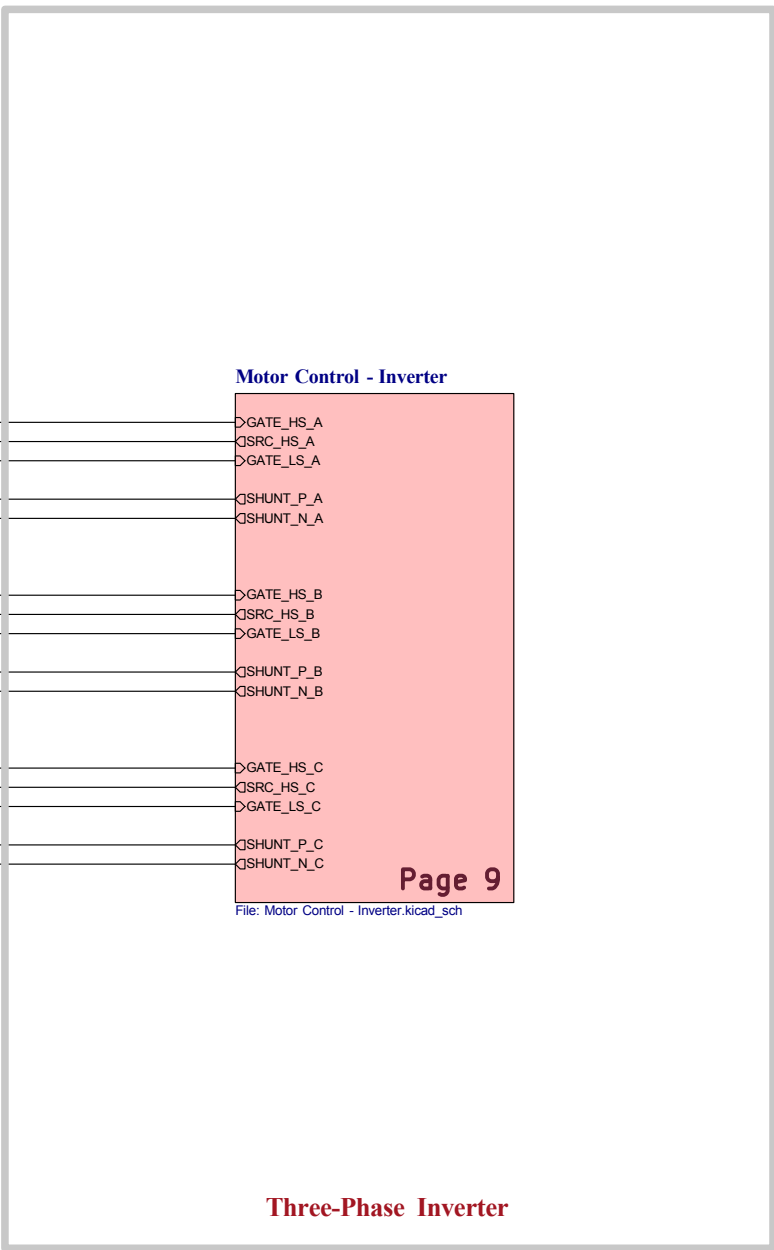
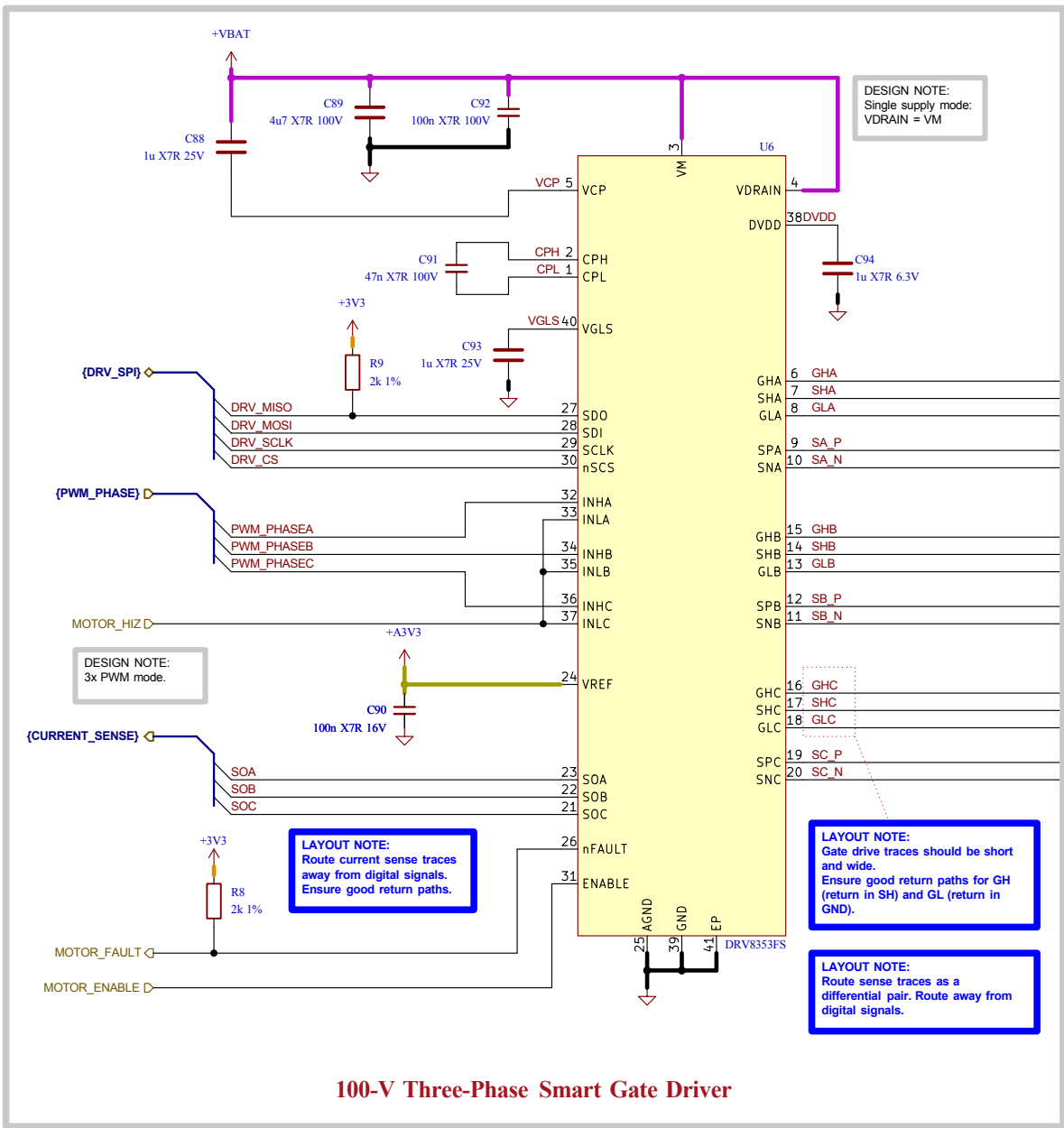
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant:		Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé		
	Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 6 of 21	

[7] Power - Connectors



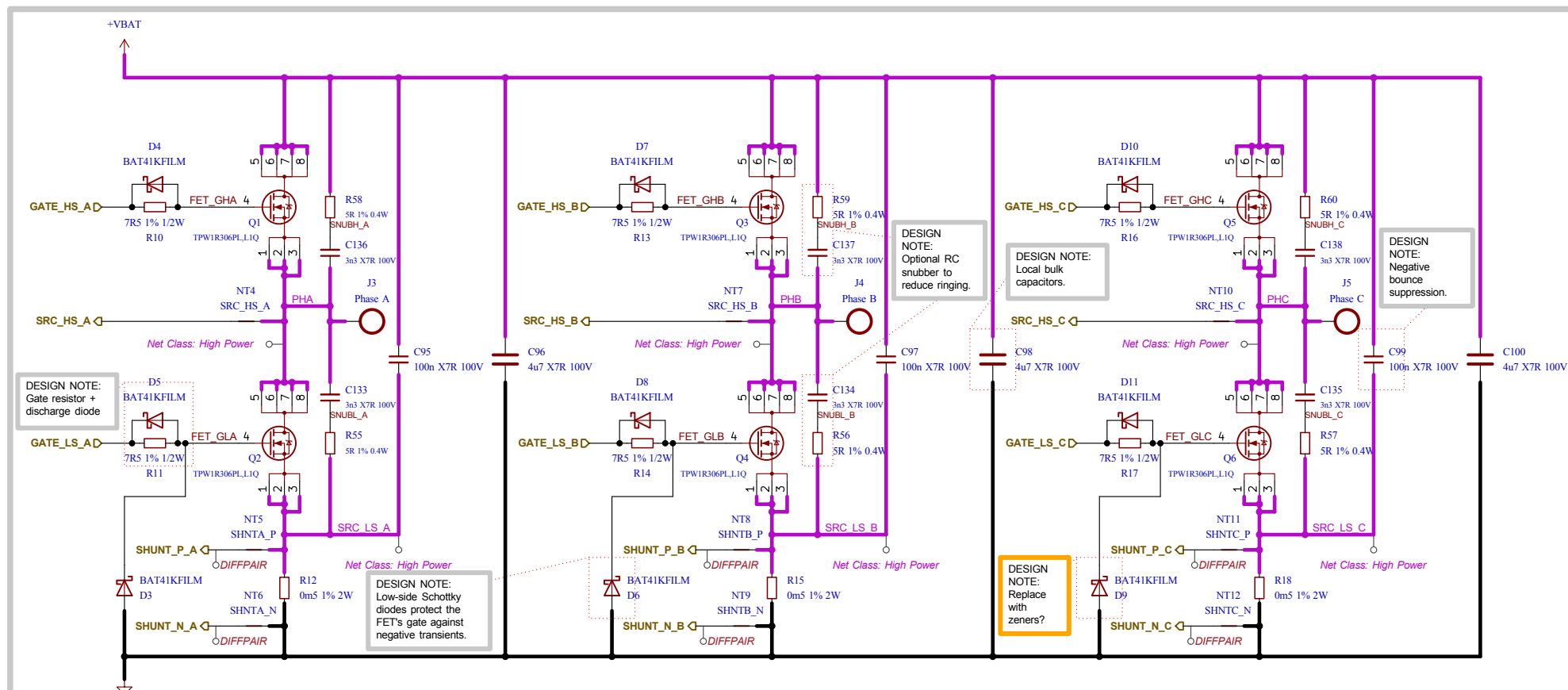
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	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Power - Connectors/	Reviewer:	Size: A4	Revision: 1.1.1+ (Unreleased)
				Sheet: 7 of 21

[8] Motor Control - Top Level



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Sheet Title: Motor Control - Top Level	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/Motor Control - Top Level/	File Name: Motor Control - Top Level.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-20	Revision: 1.1.1+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 8 of 21

[9] Motor Control - Inverter



Three-Phase Inverter

LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance
between power nets according
to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:

System Design Considerations for
High-Power Motor Driver Applications

Best Practices for Board Layout
of Motor Drivers

Proper RC Snubber Design
for Motor Drivers

Sheet Title:

Motor Control - Inverter

Sheet Path:

/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:

EPFL Xplore Research

Board Name:

Amulet Motion Controller

File Name:

Motor Control - Inverter.kicad sch

Designer:

Vincent Nguyen

Reviewer:

100

Variant:

Project Name:	
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Chienpanzé

Date:

2024-01-25

	Git Hash:
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5b68472

Enzē

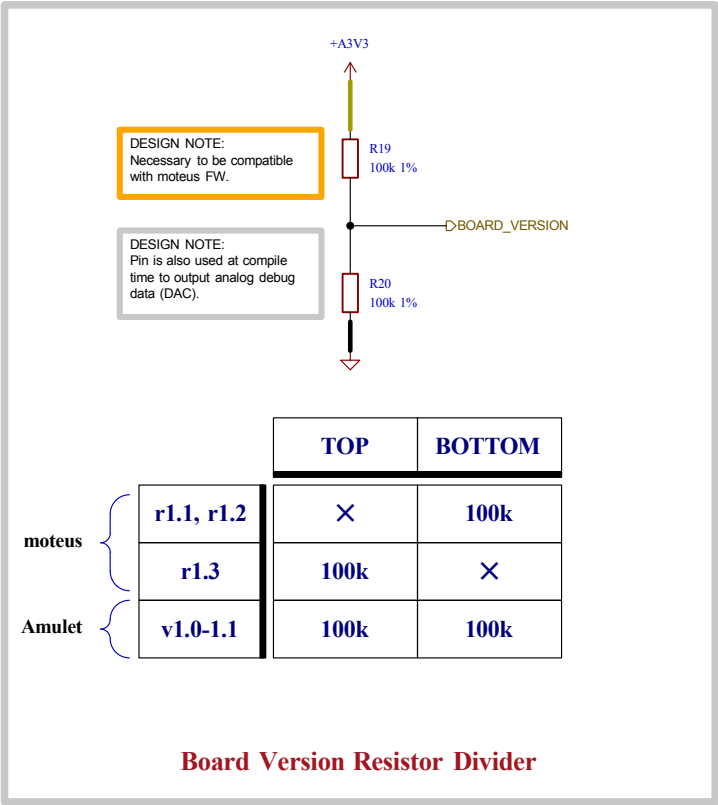
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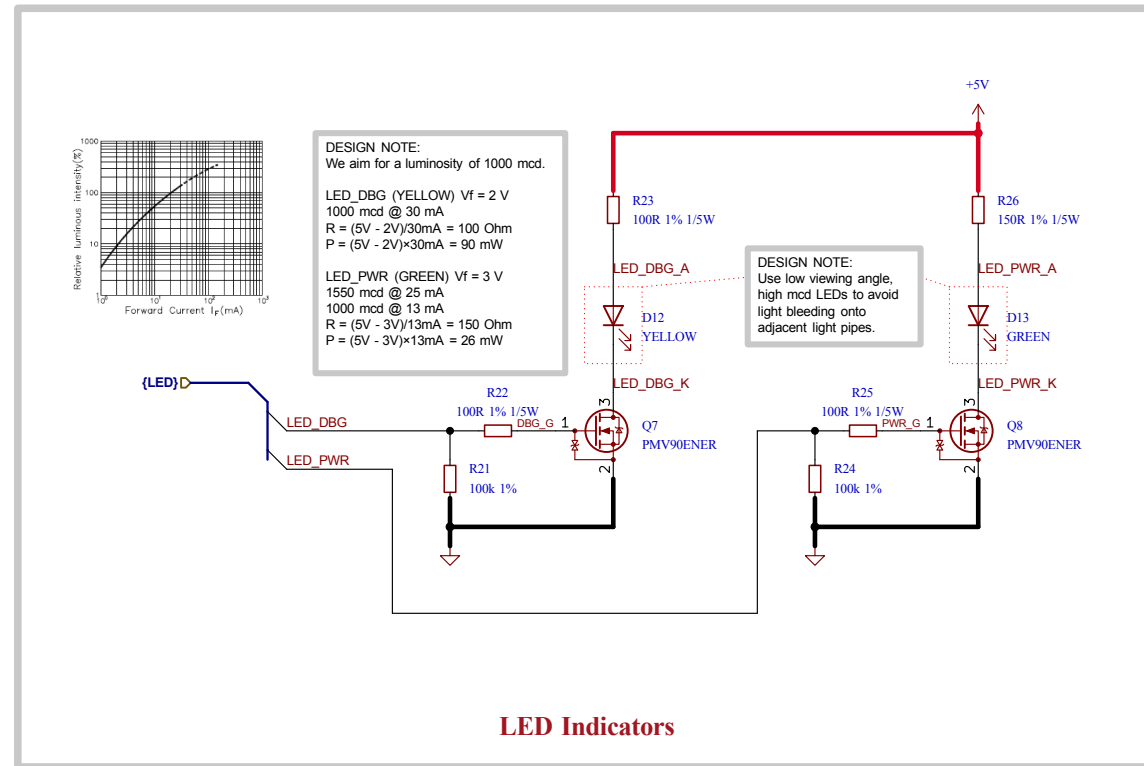
9 of 21

[10] Misc - Board Version, DAC



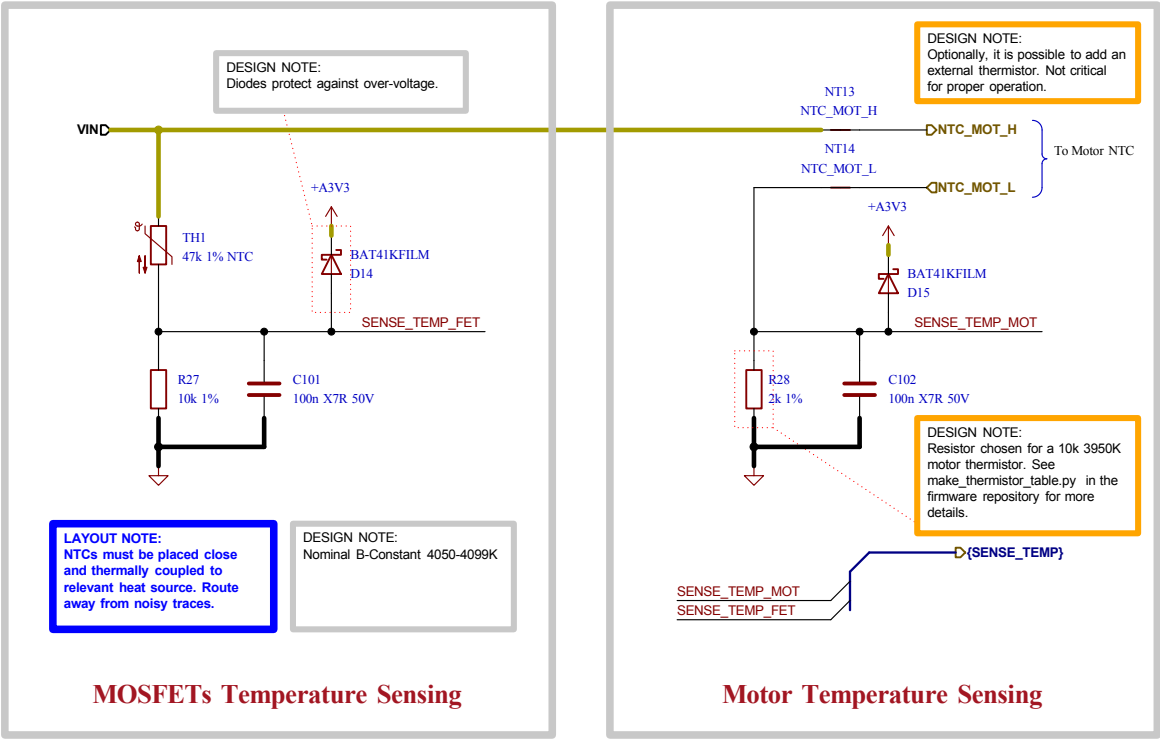
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[11] User - LED Indicators



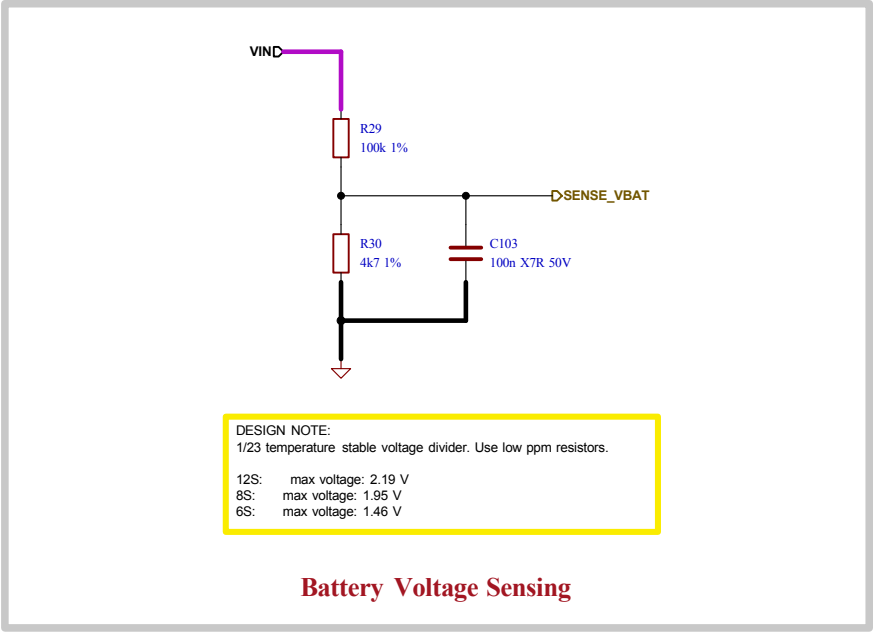
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	User - LED Indicators	Amulet Motion Controller	Chienpanzé	
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		Reviewer:	Size:	Revision:
			A4	1.1.1+ (Unreleased)
				Sheet: 11 of 21

[12] Sensing - Temperature



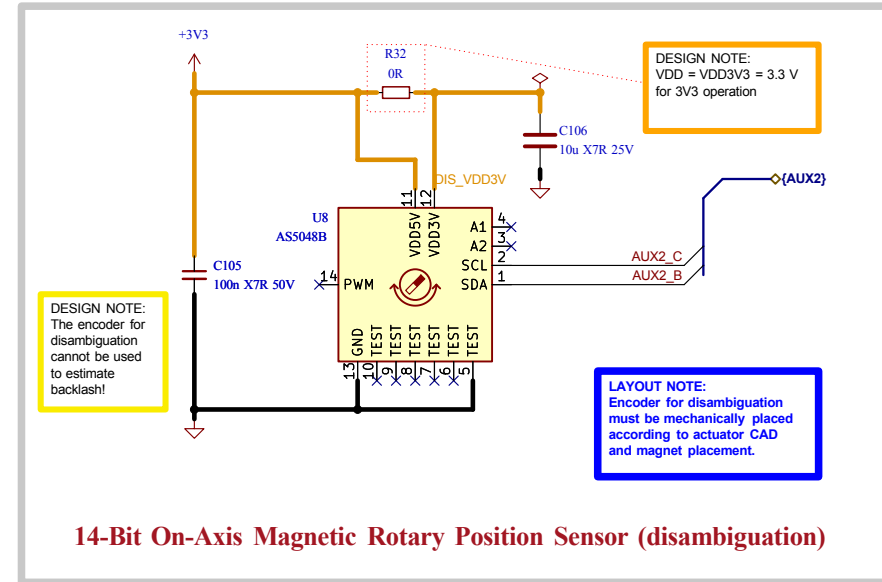
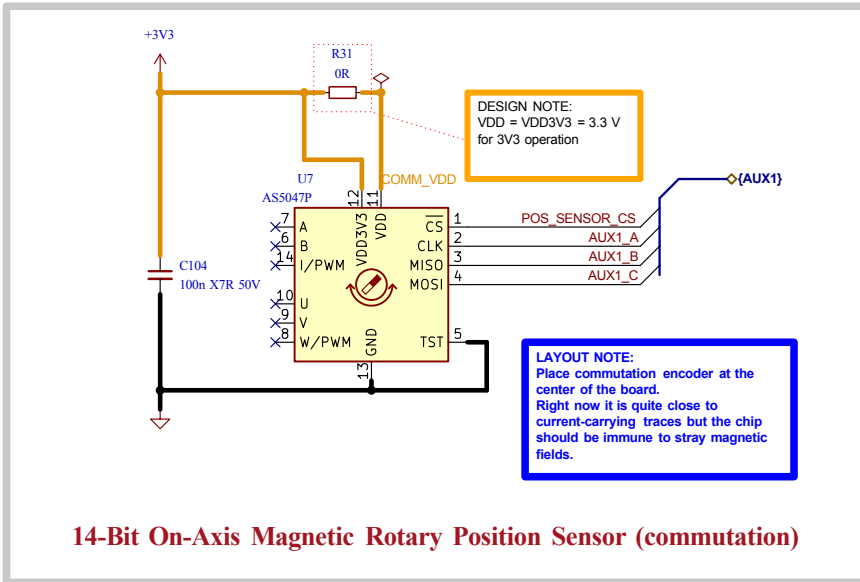
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				Sheet: 12 of 21

[13] Sensing - Battery



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
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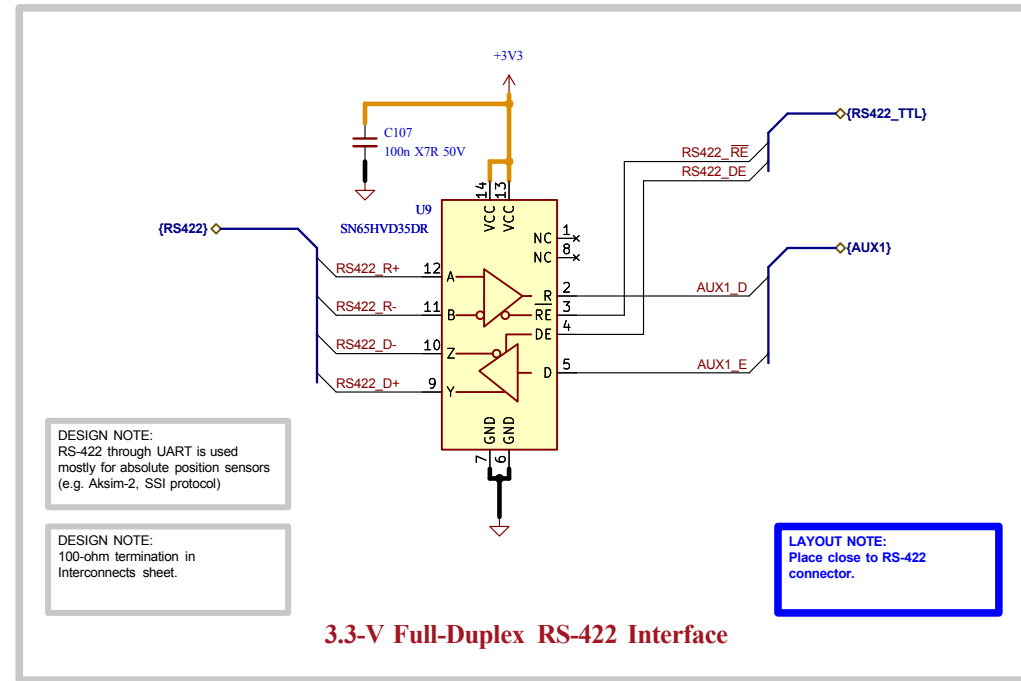
[14] Sensing - Position



DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

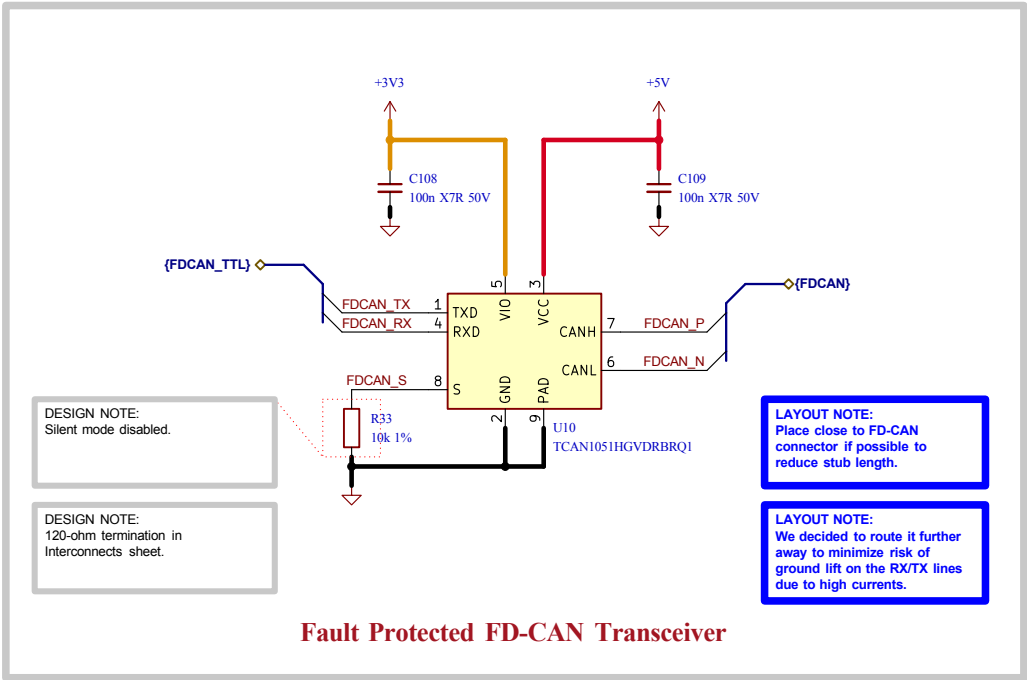
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			Reviewer:	Revision: 1.1.1+ (Unreleased)
			Size: A4	Sheet: 14 of 21

[15] Interface - RS-422



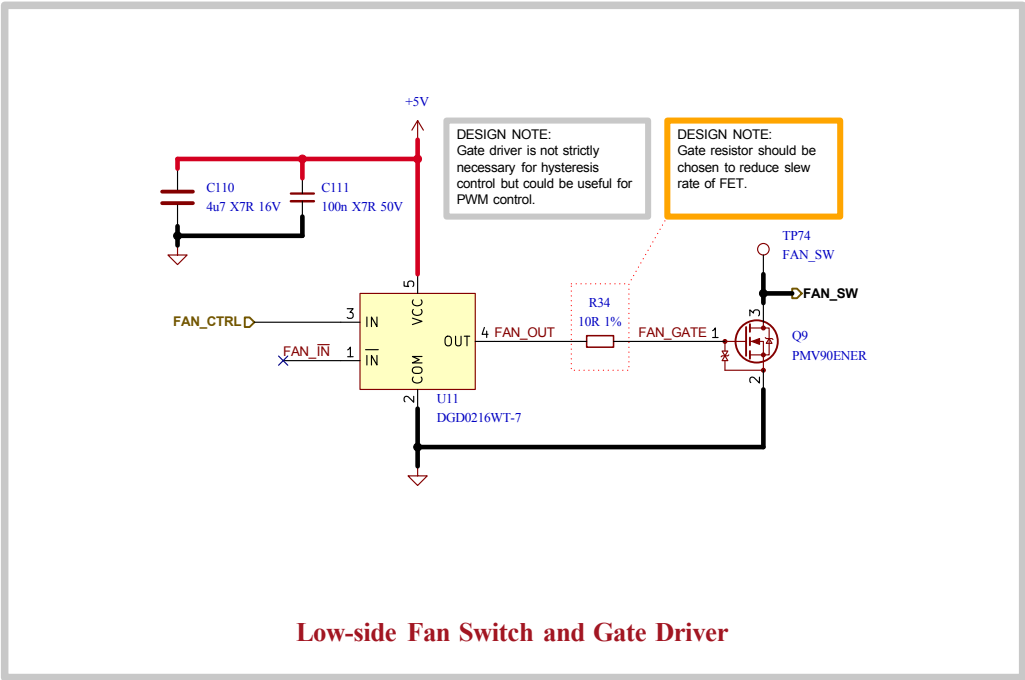
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:	Size: A4	Sheet: 15 of 21

[16] Interface - FD-CAN



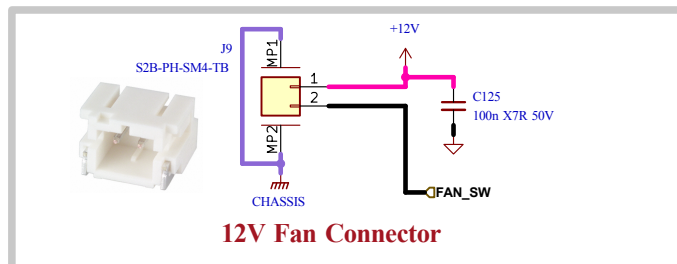
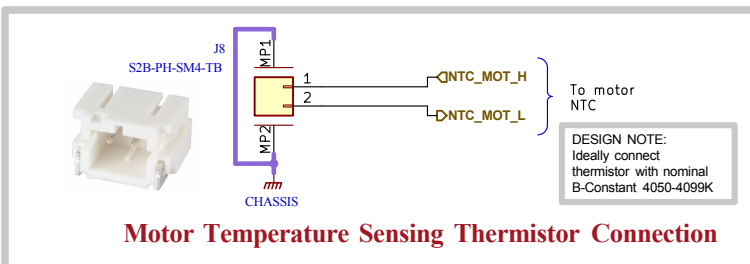
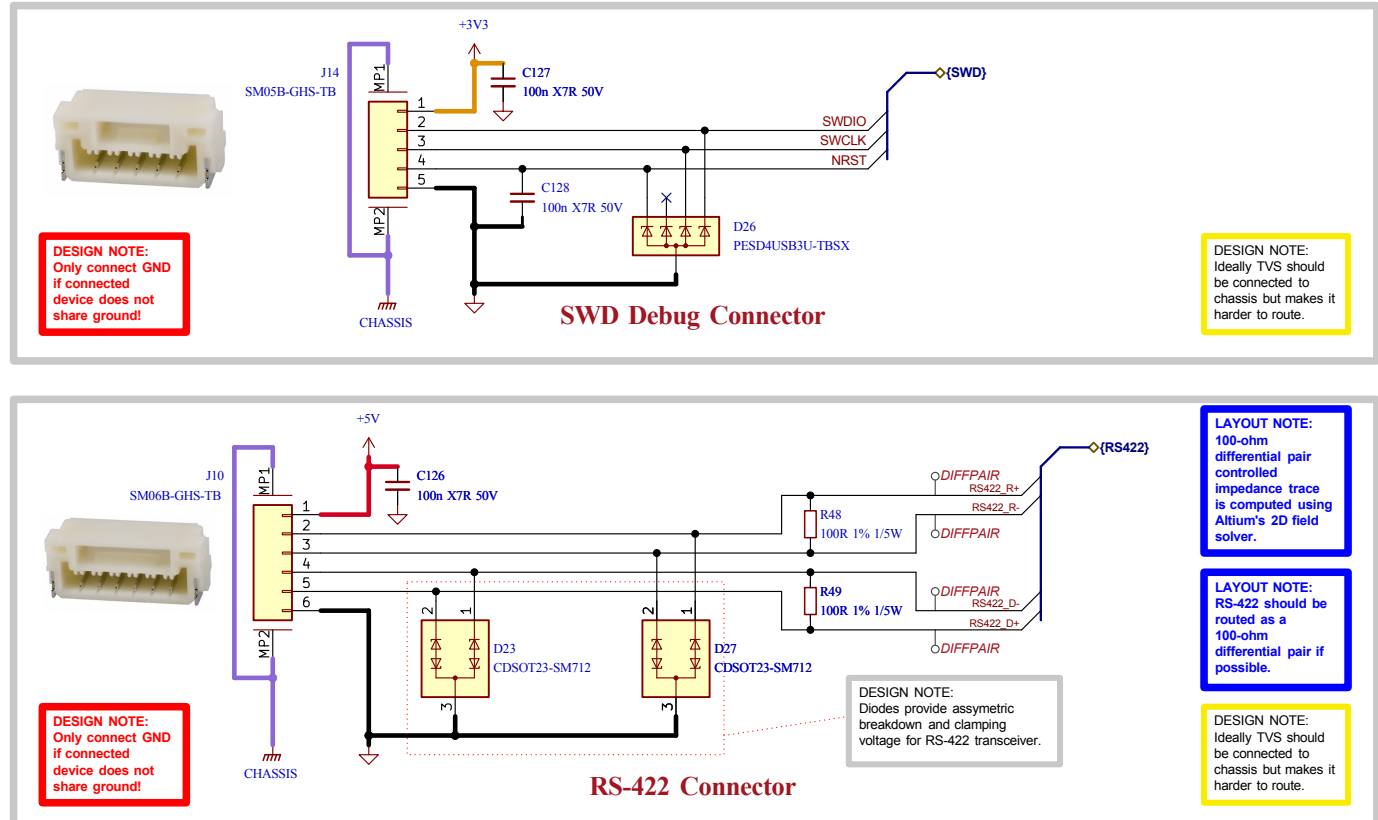
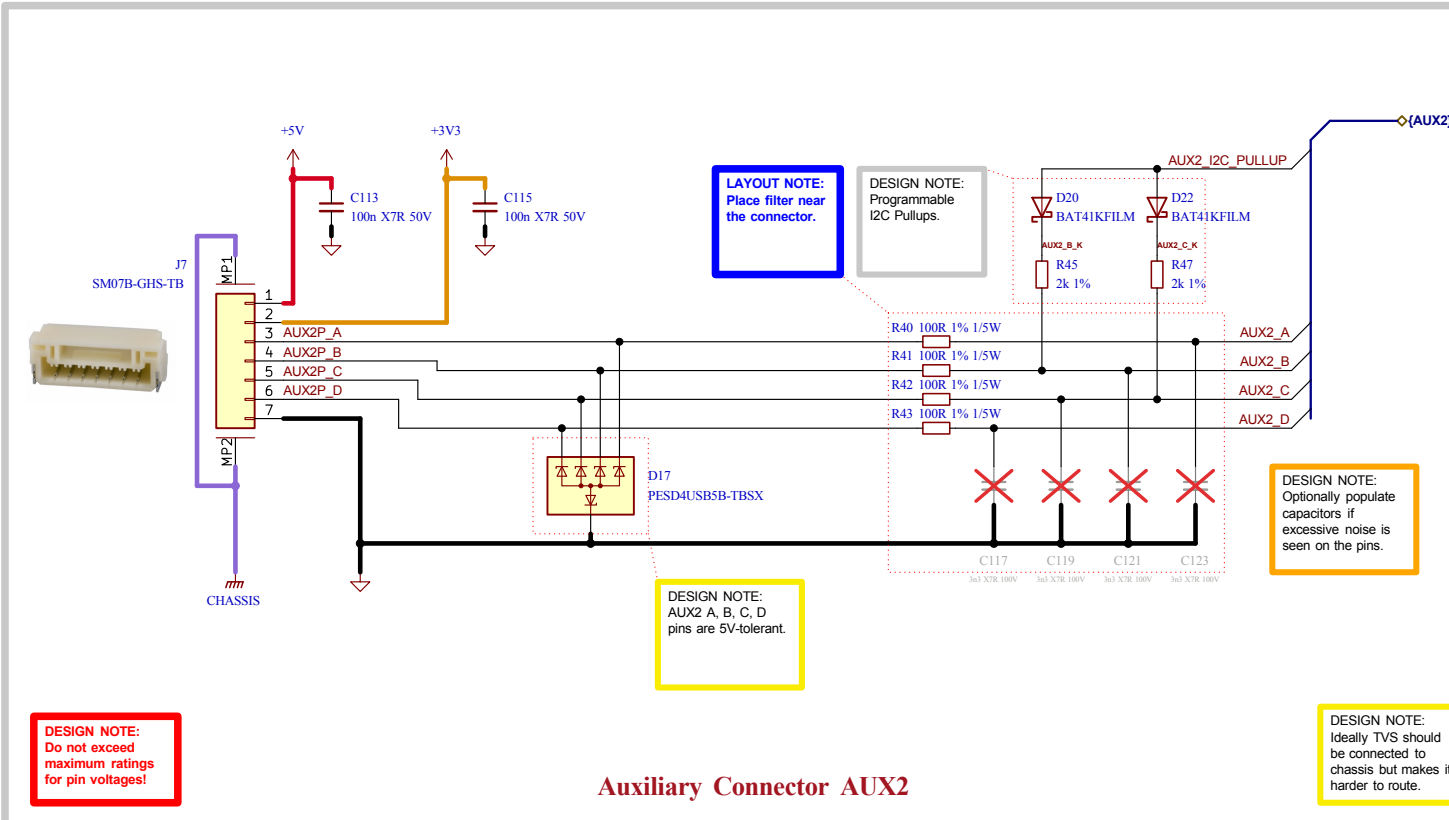
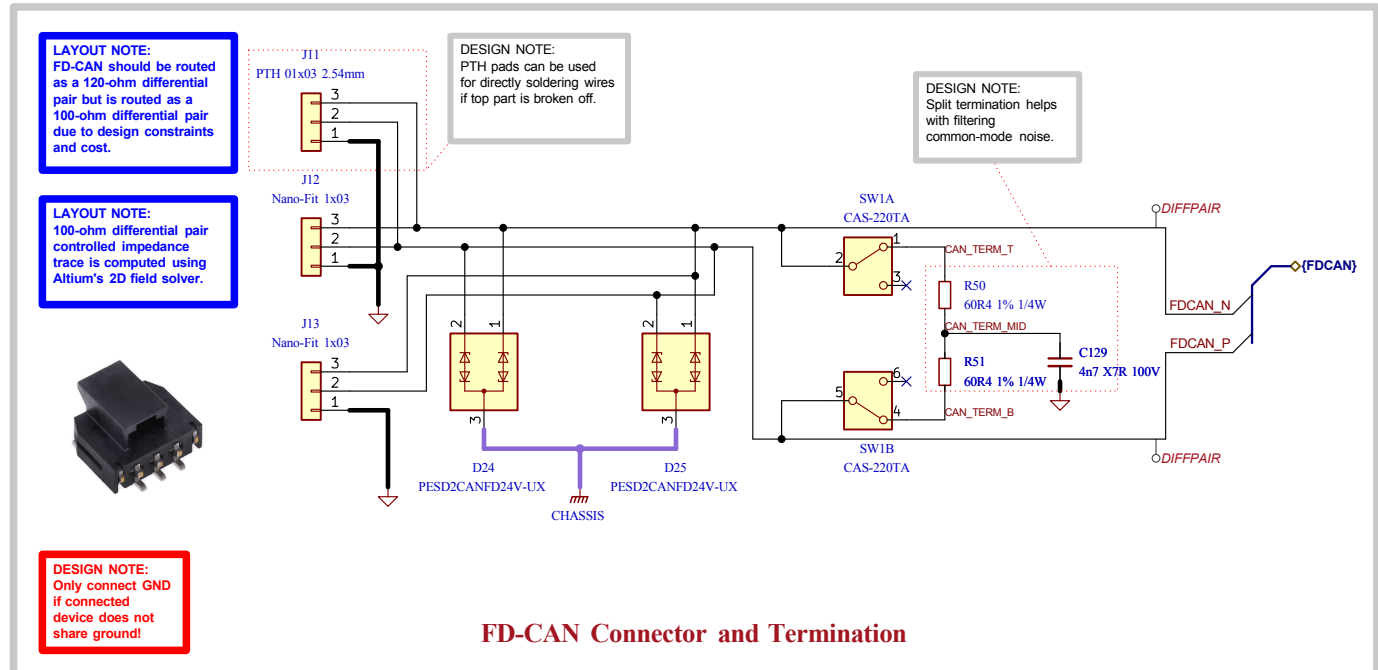
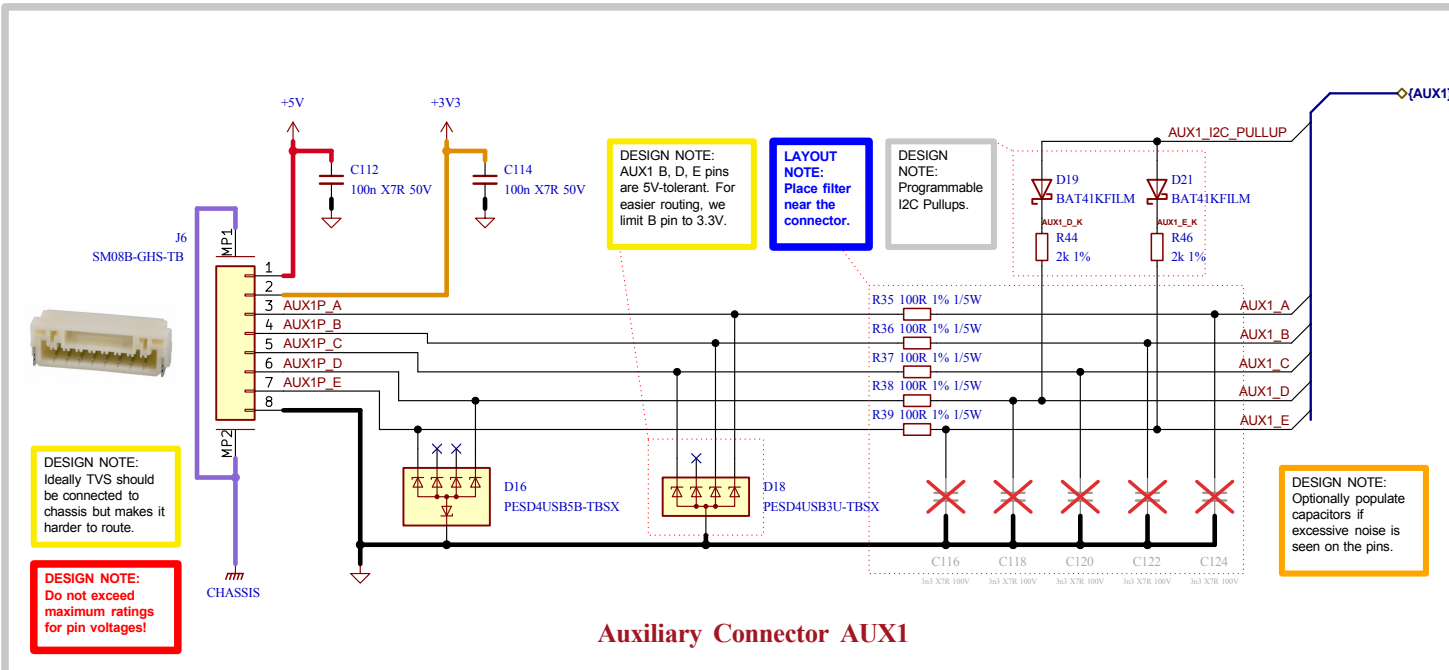
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21

[17] Interface - Fan Control



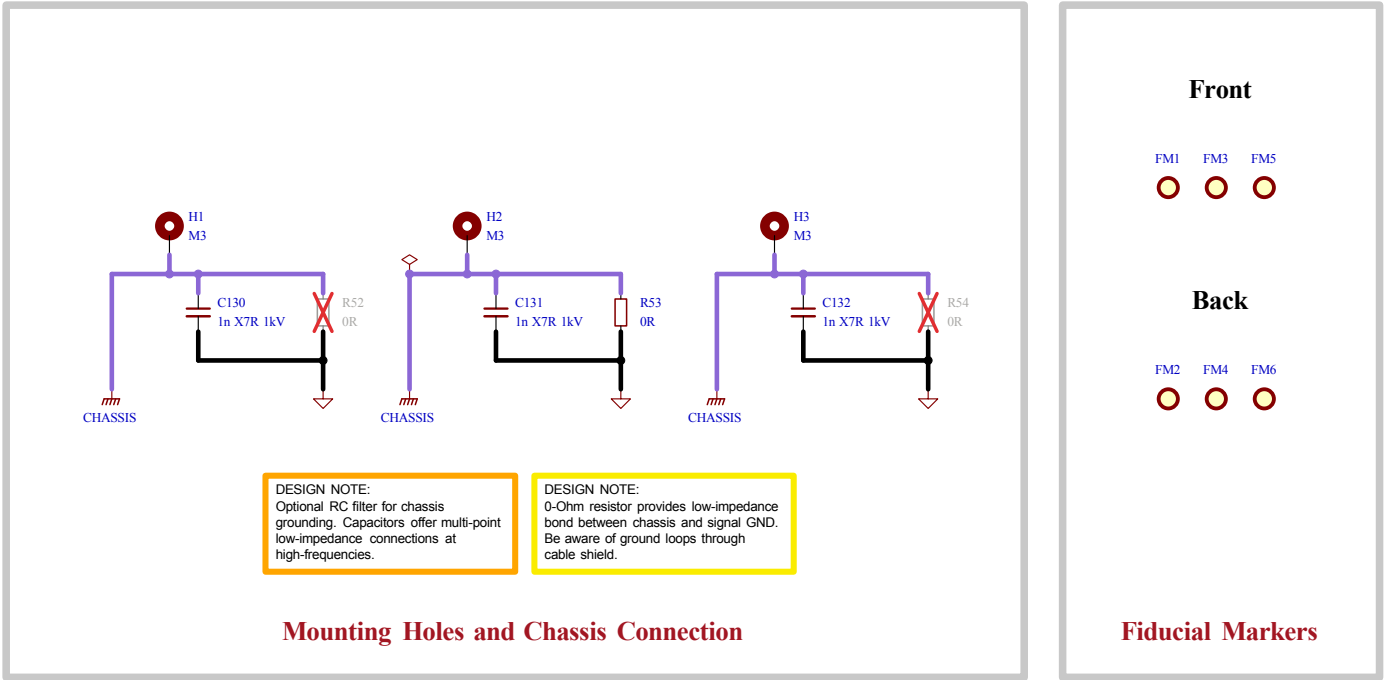
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	Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen	Date: 2023-11-19	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21

[18] Interface - Interconnects



Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



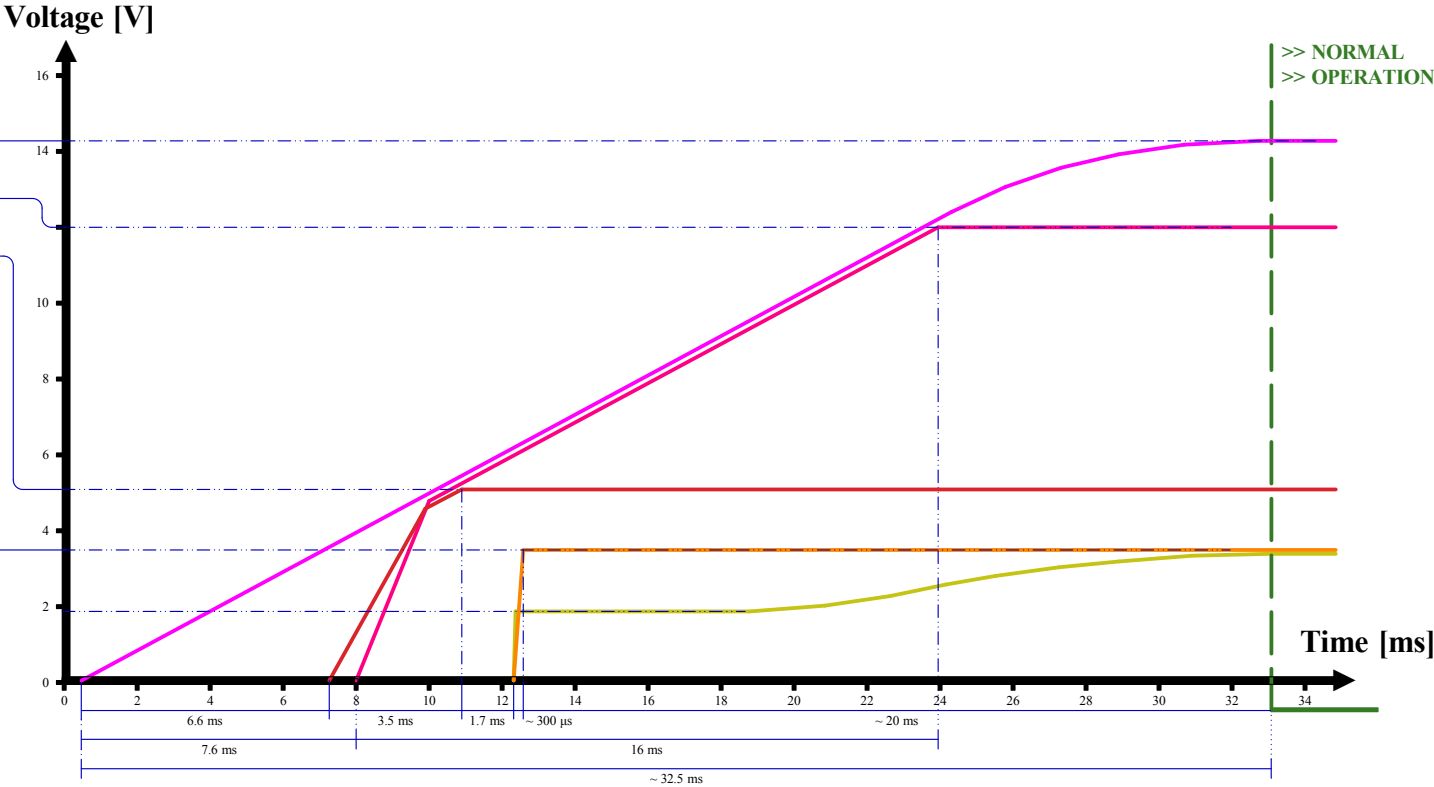
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	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21